

1. Administrative & Study Identification

Full Study Title: A multi-center, randomized, double-blind, placebo-controlled, parallel group, phase III study to evaluate the efficacy and safety of LNP023 in primary IgA nephropathy patients **Short Title/Acronym:** APPLAUSE-IgAN **Protocol Number:** CLNP023A2301 / 2020-001049-38 **NCT Number:** NCT04578834 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Iptacopan (LNP023 / Fabhalta) **Phase of Development:** Phase III **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To demonstrate the superiority of iptacopan (LNP023) compared to placebo on the reduction of proteinuria at 9 months and on slowing renal disease progression measured by the annualized total slope of estimated glomerular filtration rate (eGFR) decline over 24 months.

Secondary Objectives: To evaluate the time to first occurrence of a composite kidney failure endpoint, the proportion of participants reaching UPCR <1 g/g without initiating new therapy, the change in patient-reported fatigue (FACIT-Fatigue score), and overall safety and tolerability.

2.2 Background & Rationale To address the need for targeted therapies in IgA Nephropathy (IgAN) that slow or prevent progression to kidney failure. IgAN is a progressive autoimmune kidney disease where overactivation of the alternative complement pathway drives inflammation and kidney damage. Iptacopan is an oral, proximal complement inhibitor that selectively binds Factor B and inhibits the alternative complement pathway, thereby aiming to reduce ongoing complement-mediated injury and preserve kidney function on top of standard supportive care.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 470 adult participants (main cohort ~430 patients; severe renal impairment cohort ~40 patients) **Number of Sites:** Multicenter (Global) **Estimated Study Start:** 25-Jan-2021 **Estimated Primary Completion:** 19-Sep-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female patients ≥ 18 years of age with biopsy-confirmed IgA nephropathy. 2. For the main cohort: eGFR ≥ 30 mL/min/1.73m². For the severe renal impairment cohort: eGFR 20 to <30 mL/min/1.73m². 3. Proteinuria (UPCR ≥ 1 g/g [113 mg/mmol]) at screening and completion of the run-in period. 4. On supportive care, including a locally approved maximal daily dose or maximally tolerated stable dose of an ACE inhibitor or ARB for ≥ 90 days before study treatment.

4.2 Exclusion Criteria 1. Any secondary IgAN as defined by the investigator. 2. Patients previously treated with immunosuppressive or immunomodulatory agents (e.g., cyclophosphamide, rituximab, systemic corticosteroids >7.5 mg/d) within 90 days (180 days for rituximab) prior to first study drug. 3. History of recurrent invasive infections caused by encapsulated organisms (e.g., meningococcus and pneumococcus). 4. Active systemic bacterial, viral (including COVID-19), or fungal infection within 14 days prior to study drug administration.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Iptacopan (LNP023) 200 mg b.i.d. (twice daily) or matching placebo. **Route of Administration:** Oral. **Duration of Treatment:** 24-month treatment period.

5.2 Concomitant & Prohibited Therapy Permitted: Stable and maximally tolerated doses of background ACE inhibitors or ARBs, with or without a stable dose of SGLT2 inhibitors. **Prohibited:** Immunosuppressive therapies, systemic corticosteroids, endothelin (receptor) antagonists within 90 days, or oral budesonide.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening & Run-in	Day -90 to Day -1	Informed Consent, Biopsy Confirmation, UPCR & eGFR Labs, Stabilization of Background Therapy
Baseline	Day 1	Randomization (1:1), First Dose of Iptacopan or Placebo
Interim	Month 9	Efficacy Assessment (Proteinuria/UPCR evaluation), Safety Labs, FACIT-Fatigue review
End of Study	Month 24	Final eGFR Assessment, Final Safety & Efficacy Evaluation, IP Accountability

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: 1) Reduction of proteinuria at 9 months (Interim Analysis). 2) Annualized total slope of estimated

glomerular filtration rate (eGFR) decline over 24 months (Final Analysis). **Measurement Method:** 24-hour urine collection for Urine Protein to Creatinine Ratio (UPCR) and standardized central laboratory assessment for serum creatinine/eGFR.

7.2 Secondary & Exploratory Endpoints 1. Time from randomization to the first occurrence of a composite kidney failure endpoint (sustained $\geq 30\%$ decline in eGFR, eGFR <15 mL/min/1.73m², dialysis, kidney transplant, or death from kidney failure). 2. Proportion of participants reaching UPCR < 1 g/g without initiating new background therapy or immunosuppressants. 3. Change from baseline to 9 months in patient-reported fatigue via the FACIT-Fatigue questionnaire.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via site visits and laboratory assessments, focusing specifically on infections (e.g., encapsulated bacterial infections, COVID-19, URTI), headache, and hyperlipidemia. **Serious Adverse Events (SAEs):** Reported within standard 24-hour global safety timelines to the sponsor. **Data Monitoring Committee (DMC):** An independent committee oversees the pre-specified interim analysis at 9 months and conducts routine surveillance of patient safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Main efficacy population (eGFR ≥ 30 mL/min/1.73m²); severe renal impairment cohort analyzed separately; Safety Set for all participants receiving at least one dose. **Sample Size Justification:** Approximately 470 patients (430 main cohort, up to 40 severe renal impairment) powered at $\alpha = 0.05$ to detect superiority of iptacopan over placebo in UPCR reduction and eGFR slope. **Handling of Missing Data:** Utilizing Mixed Models for Repeated Measures (MMRM) or multiple imputation for longitudinal eGFR and UPCR evaluations.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring visits to ensure protocol adherence, source data verification, and drug accountability. **Audit Trail:** eCRF entry tracking, robust data clarification workflows, and strict compliance with required vaccinations prior to Factor B inhibition.

11. Study Identifiers & Governance

Primary ID: CLNP023A2301 **Registry Number:** NCT04578834 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** APPLAUSE-IgAN **Official Regulatory Title:** A multi-center, randomized, double-blind, placebo-controlled, parallel group, phase III study to evaluate the efficacy and safety of LNP023 in primary IgA nephropathy patients

12. Clinical Framework (IgA Nephropathy Specific)

Primary Condition: Primary IgA Nephropathy (IgAN) **Diagnosis Threshold:** Biopsy-confirmed IgAN with UPCR ≥ 1 g/g **Medical Methodology:** Alternative Complement Pathway (Factor B) Inhibition **Optimization Target:** Proteinuria reduction and preservation of eGFR slope

13. Study Procedures & Methodology

Management Pathway: Standard supportive care utilizing maximally tolerated ACEi or ARB **Education Protocol (HCP):** Site initiation and training on encapsulated bacterial infection risks, mandatory vaccination screening, and specialized complement-pathway monitoring. **Education Protocol (Patient):** Informed consent process detailing the necessity of vaccinations against *Neisseria meningitidis*, *Streptococcus pneumoniae*, and *Haemophilus influenzae* prior to dosing. **Quality Audit Mechanism:** Central laboratory review of baseline biopsy eligibility and compliance monitoring regarding background therapy stabilization.

14. Observation & Follow-Up Schedule

Total Duration: 24-month treatment period (plus up to 3 months screening/run-in) **Onsite Visit Cadence:** Regular intervals including Screening, Day 1, Month 9 (interim analysis), and Month 24 (End of Study). **Remote Monitoring Frequency:** Continuous monitoring per standard clinical operational plans and regulatory requirements. **Checklist Review Interval:** Regular evaluations of concomitant medications to ensure stabilization of background therapy.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]				
Statistician	[Name]	____	[DD-MMM-YYYY]				